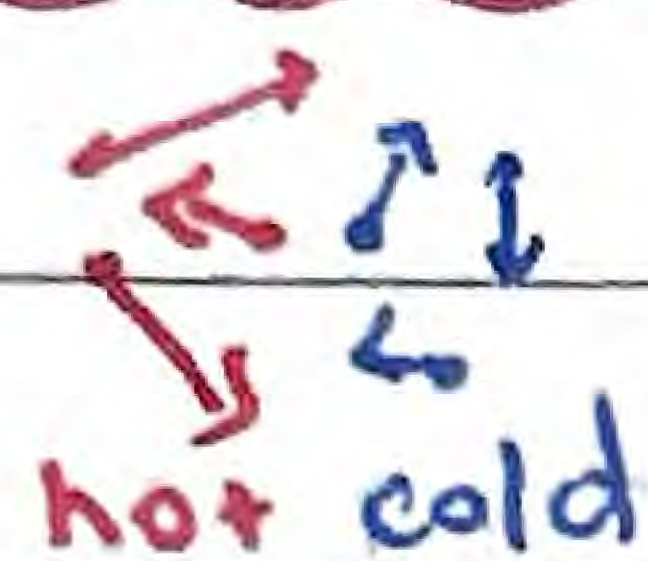


Energy Emission

Recall energy transfer occurs from

1. Conduction - transfer of energy through collisions between neighboring atoms or molecules



2. Convection - transfer of energy by movement of a heated fluid in a cyclical manner



↳ Via molecular flow

3. Radiation - transfer of energy through electromagnetic waves from a hot body to a colder one

Energy commonly refers to that of mechanical, thermal, electric, etc.

Rate of energy transfer is given by power, $P = \frac{\Delta E}{\Delta t}$

Intensity - Amount of energy passing through a unit area per unit time, or power per unit area

↳ Rate of energy transfer perpendicular to the direction of propagation

$$I = \frac{P}{A} = \frac{E}{A \cdot t}$$

for...

I - intensity

E - energy

P - power

t - time

A - area

given in $W \cdot m^{-2}$

Blackbody Radiation

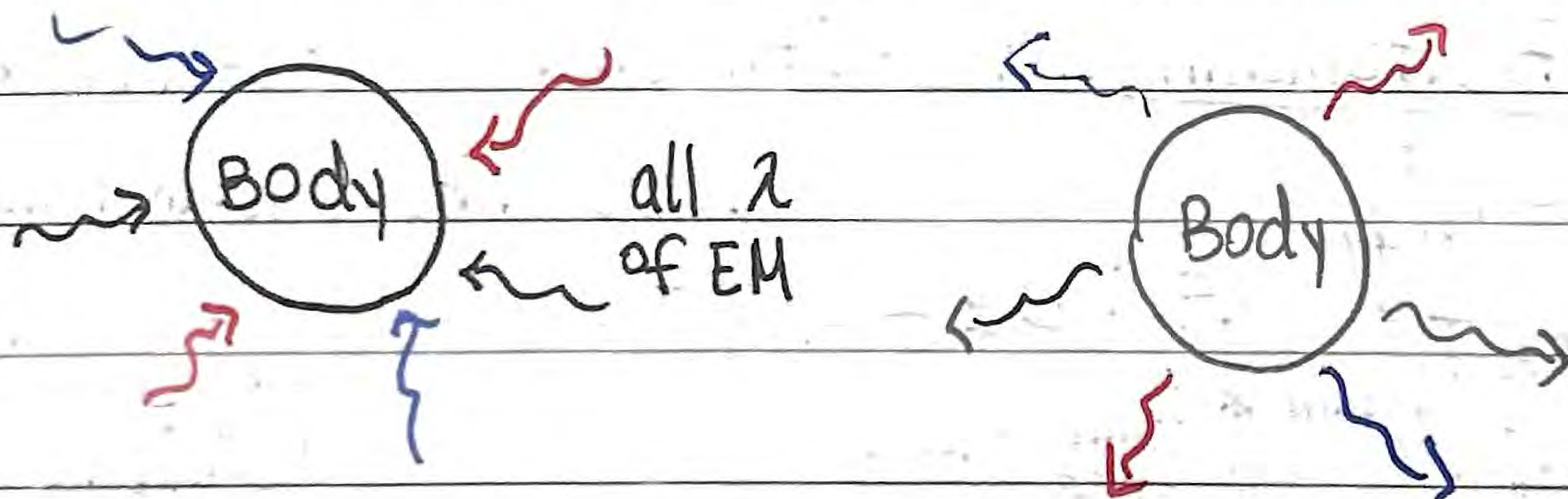
Refers to the thermal radiation emitted by all bodies

↳ Can be infrared, or visible, etc. depending on temperature

A perfect black body, one that absorbs/emits blackbody radiation, is defined as...

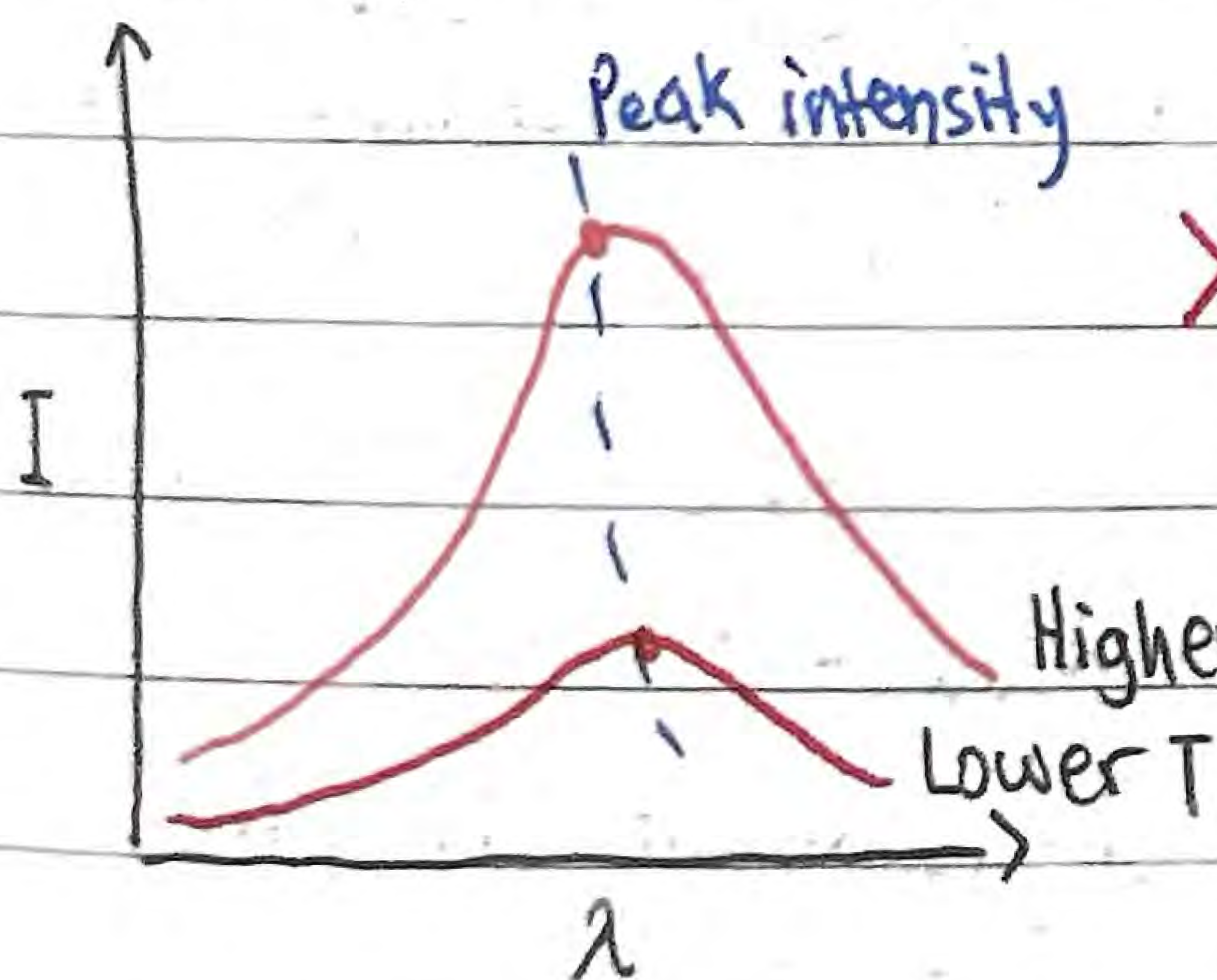
An object that absorbs all of the radiation incident on it and does not reflect nor transmit any

It's a perfect absorber, hence also perfect emitter



Absorption of all λ makes body appear black

Intensity and wavelength distribution of any emitted wave depend on temp.



> As $T \uparrow$, $I_{\text{peak}} \uparrow$. But $\lambda_{\text{peak}} \downarrow$
(λ shift left)

Higher T ↳ This indicates temperature
causes a change in color

(as λ with highest intensity is perceived)

Blackbody Radiation

Apparent Brightness - Intensity of radiation on Earth received from a star

↳ Depends on how much light the star emits, how far the star is

↳ Tells us brightness as measured by observer

Luminosity - Total power output of radiation emitted by a star

↳ Total power output of a star, brightness at its surface

Hence, apparent brightness (intensity) is typically a fraction of luminosity

Stefan-Boltzmann Law - Total radiated power of a black body

depends on its absolute temp. and surface area

$$P_{\text{emit}} = \sigma A T_{\text{body}}^4$$

$$P_{\text{absorb}} = \sigma A T_{\text{surround}}^4$$

Where...

$$P = \sigma A T^4$$

$$P_{\text{net}} = e \sigma A (AT)^4$$

↳ when object is not in equilibrium (temp. in change)

P - Total power absorbed/emitted across all wavelengths; A - surface area of body

σ - Stefan-Boltzmann constant ($5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$)

T - Absolute temperature of surrounding/body

$e = \frac{\text{power radiated by object}}{\text{power emitted by black body}}$

no emission (e.g. black hole)

perfect black body (e.g. star)

$L = \text{luminosity} = P = e \sigma A T^4$ for $e \in (0, 1)$, emissivity of surface

useful since planets, unlike stars, are not approximate to black bodies

Wien's displacement law - relates observed λ to surface temperature

$$\lambda_{\text{max}} \cdot T = 2.9 \times 10^{-3} \text{ m K}$$

for $\lambda_{\text{max}} \propto \frac{1}{T}$

where λ_{max} - greatest intensity λ

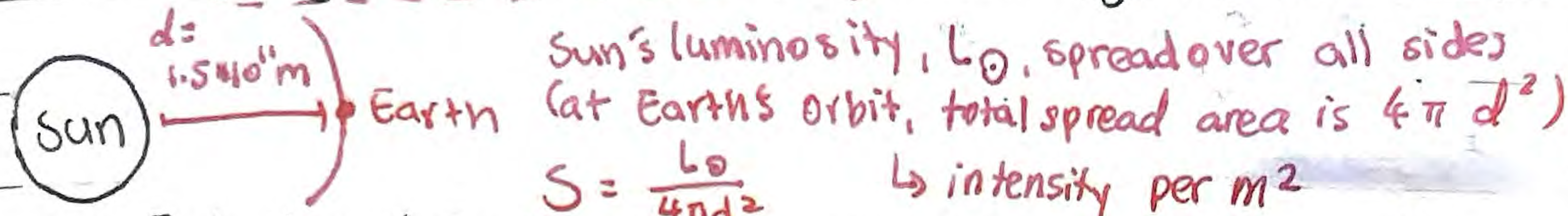
T - surface temperature

↳ As temp. $\uparrow$, λ emitted $\downarrow$

Energy Balance Model of the Planet

Solar Constant - Intensity of Sun's radiation arriving perpendicularly to the Earth's position at average orbit radius

↳ $S_{avg} = 1.36 \times 10^3 \text{ W m}^{-2}$ (but changes because of elliptical orbit)



$$S = \frac{L_{\odot}}{4\pi d^2} \quad \rightarrow \text{intensity per m}^2$$

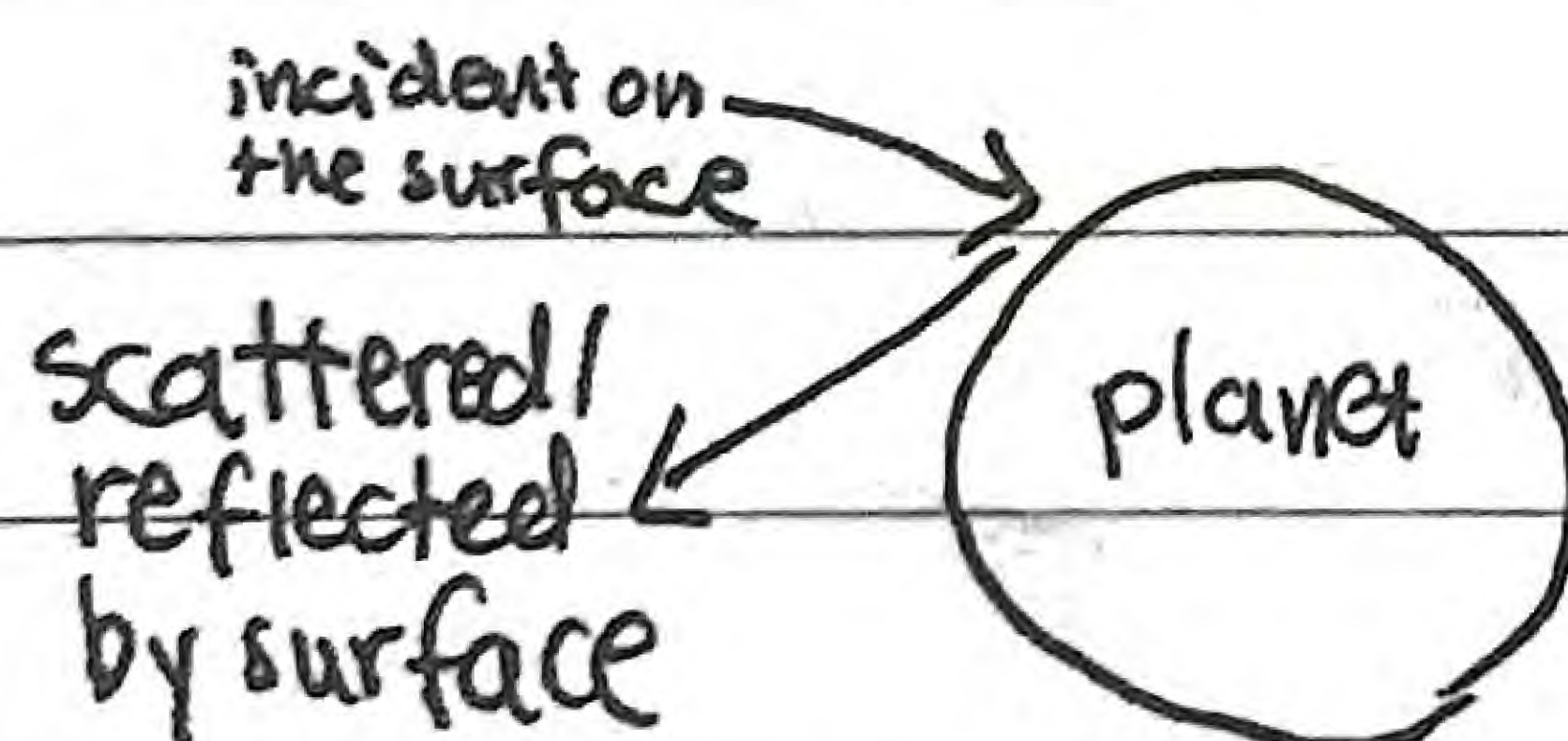
For Earth, it captures intensity over circular cross section, spreads it over entire surface: $(S \times \pi r_{\text{earth}}^2 \div 4\pi r_{\text{earth}}^2) \Rightarrow \frac{S}{4}$ (mean solar intensity at planet surface)¹⁰

* But we need to quantify ability of a non-blackbody to:
 { absorb, reflect, scatter thermal radiation
 { emit thermal radiation

$$\alpha = \text{albedo} = \frac{\text{total power scattered}}{\text{total power incident}} = \frac{\text{total intensity scattered}}{\text{total intensity incident}}$$

consequently... $\alpha = 1 - \frac{\text{total absorbed}}{\text{total incident}}$

Where...



* Scatter ≠ emission
 ↓ ↓
 reflected off acts as source
 (scatter happens before absorbed)

$0 < \alpha < 1$ when all radiation is scattered
 when all power is absorbed

E.g.

Material

Albedo

ocean water

0.1

forest

0.1

ocean ice

0.4

.....

...

Energy Balance Model of the Planet

E.g. Radiation of intensity 610 W m^{-2} was incident perpendicular on the surface of a lake

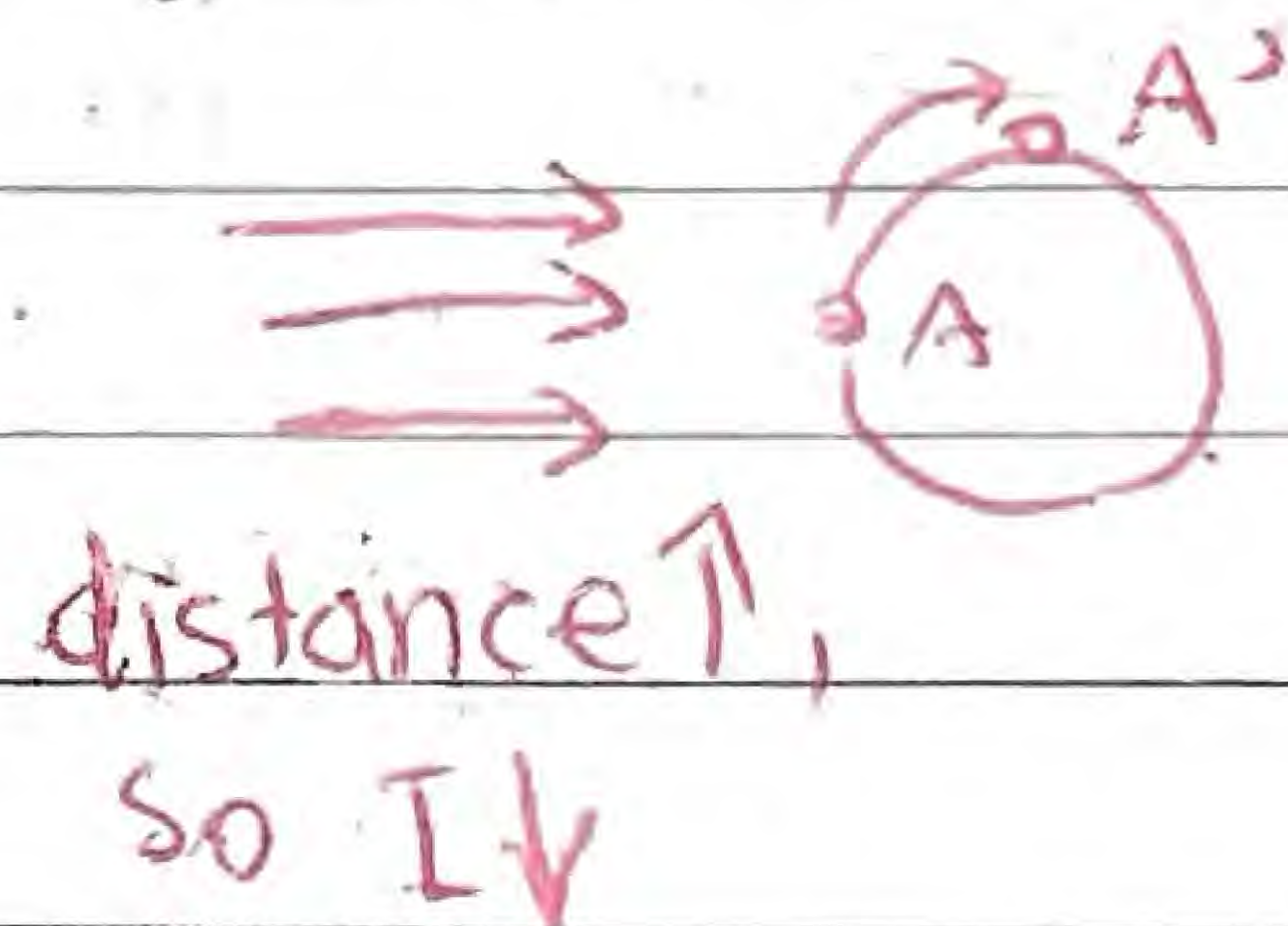
a) If the lake has an albedo of 0.18, calculate how much intensity was absorbed by the surface

$$I_{\text{absorbed}} = 610 (1 - \alpha)$$

$$= 610 (1 - 0.18) = \underline{500 \text{ W m}^{-2}}$$

b) Describe how answer in (a) would change later in the day

I_{absorbed} decreases :

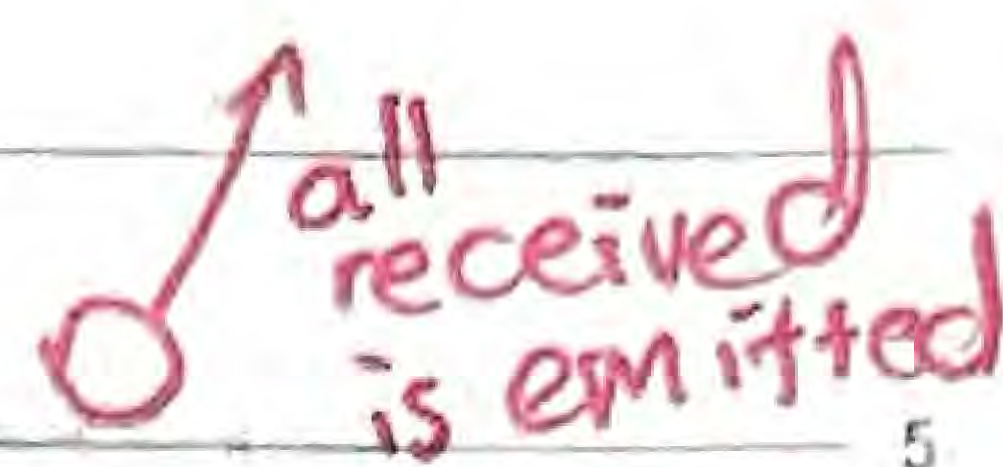
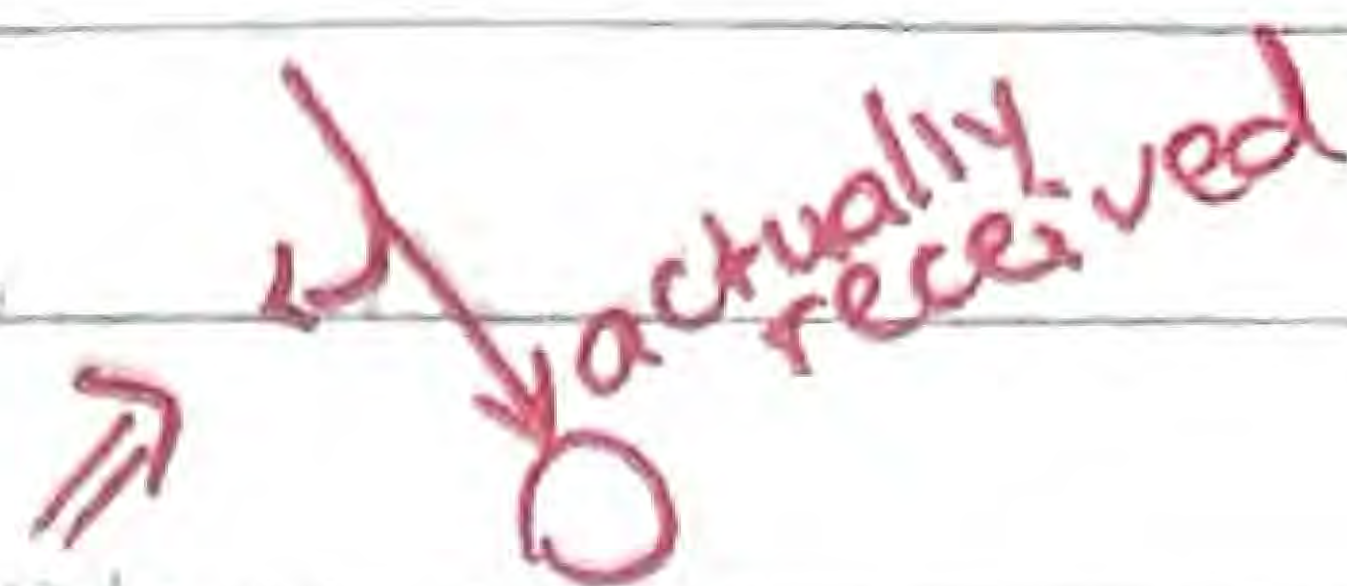


Model assumption:

For a planet with no atmosphere in thermal equilibrium the outgoing radiated intensity equals the absorbed intensity

Energy Balance Model of the Planet

The IB uses the following:



> Radiant thermal energy = Radiant thermal energy
 received by planet emitted by planet
 (with albedo considered)
 both under same time interval

this is also energy absorbed by planet

or...

$$\frac{S}{4} (1 - \alpha) = e \sigma T^4$$

← emissivity (with respect to black body)

← albedo (with respect to body)

In the case of the Earth...

$$\frac{S}{4} (1 - \alpha) = e \sigma T^4$$

$$\frac{1.36 \times 10^3}{4} (1 - 0.315) = (0.61) (5.67 \times 10^{-8}) T^4$$

$$T = 286 \text{ K} = 13^\circ \text{C} \quad \leftarrow \text{Average temperature of the Earth.}$$

Although model is highly simplified, it's reasonably accurate in concluding that:

1. The values of emissivity and albedo have a fundamental effect on the planet's temperature
2. The values of e and α are being changed by human activity
3. Small changes in the Earth's temperature may produce large changes in the Earth's climate.
 ↳ Such changes are harmful to our lives

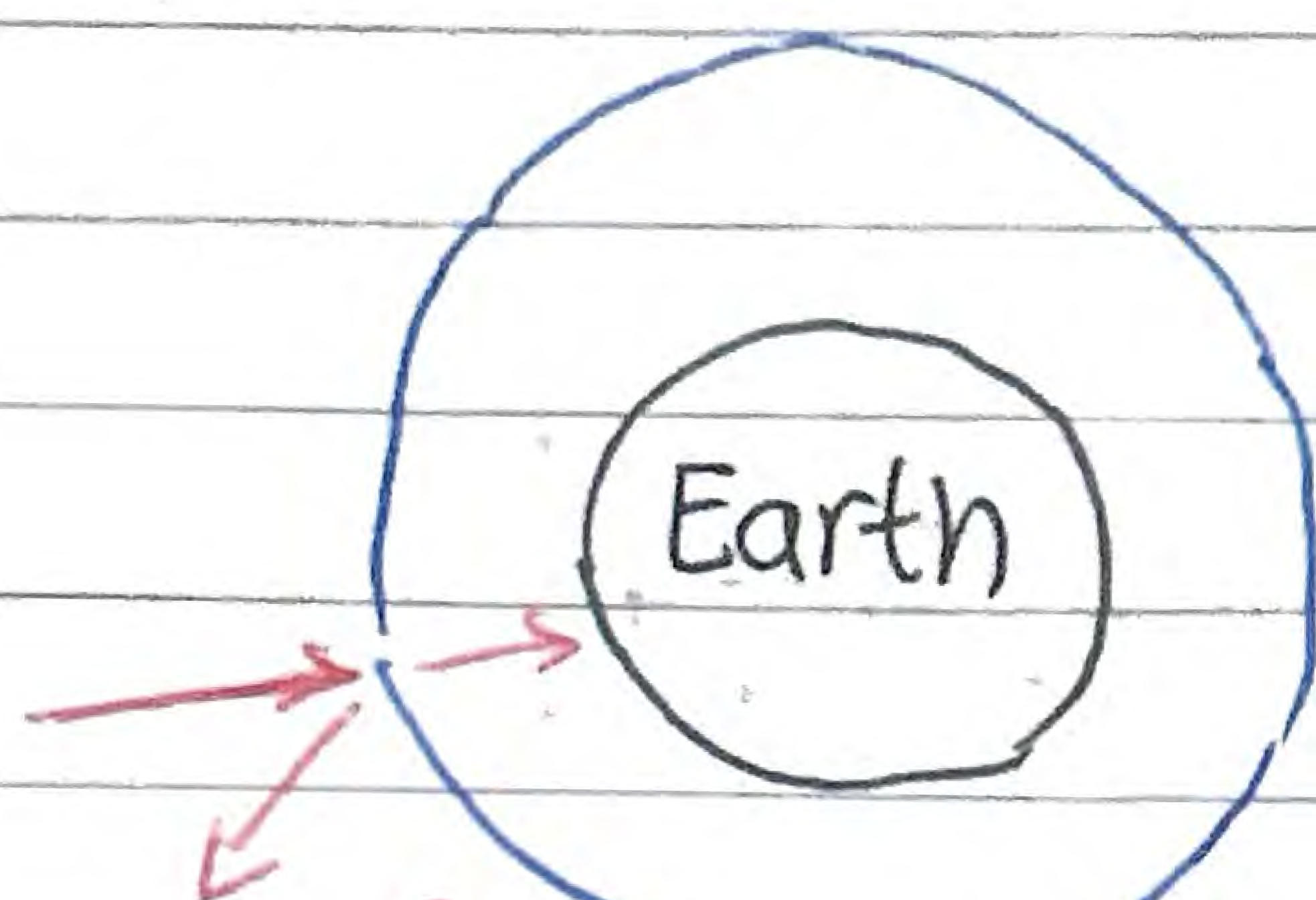
Greenhouse Effect

IB's model reveals that human activity causes temperature rises on the Earth's surface

• Greenhouse effect furthers this idea

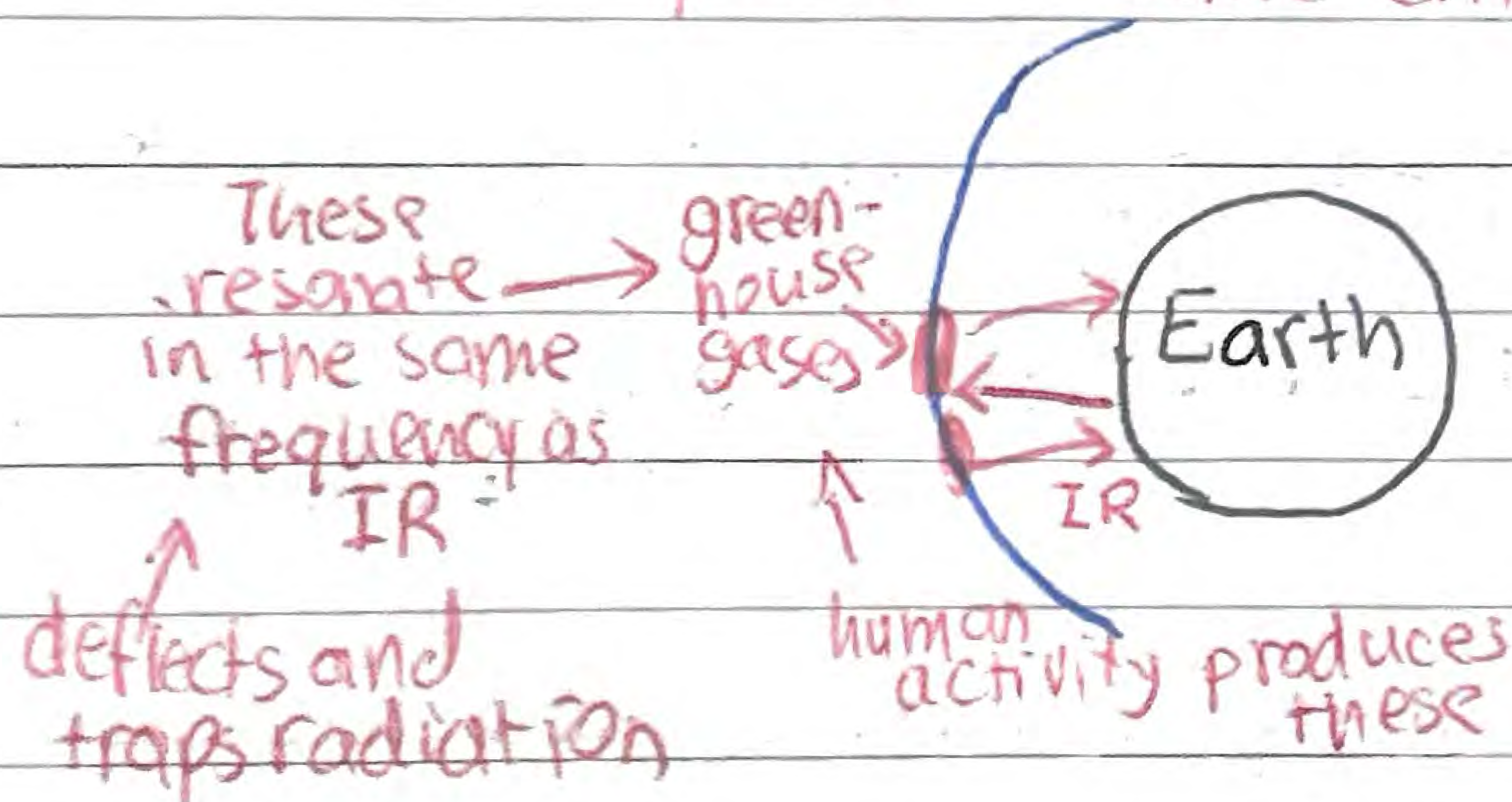
(adds to simplified model)

since emission can lead to more absorption from reradiated energy)



outer space radiation

hits atmosphere $\Rightarrow$ some are deflected,
some enter



Greenhouse Effect - Physical mechanism by which some of the radiation (IR) emitted by the surface of the planet is absorbed by the greenhouse gases in the atmosphere and re-radiated towards the surface

Greenhouse Effect

Essentially, gases in our atmosphere trap energy and release it into the atmosphere

↳ This causes atmosphere to warm

Main Greenhouse Gases include...

Sources:

natural

human

> Methane

wetlands

agriculture

> Water vapor

clouds, oceans

fossil fuel
combustion

> Carbon dioxide

plant respiration,
surface ocean

fossil fuel
combustion

> nitrous oxide

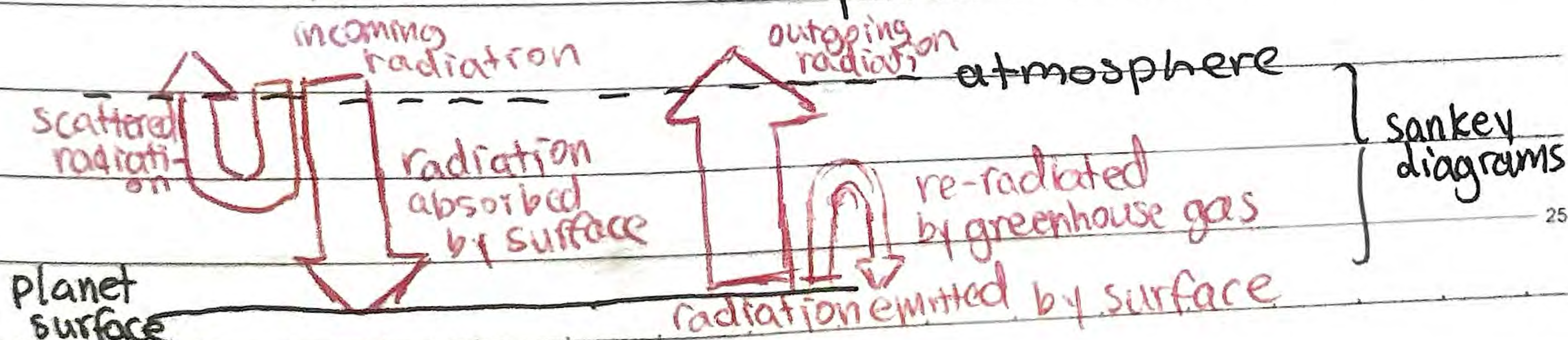
forests, oceans

agriculture

Main steps

1. Absorption of radiation causes resonance with greenhouse gas molecules (radiation matches energy difference between molecular levels)

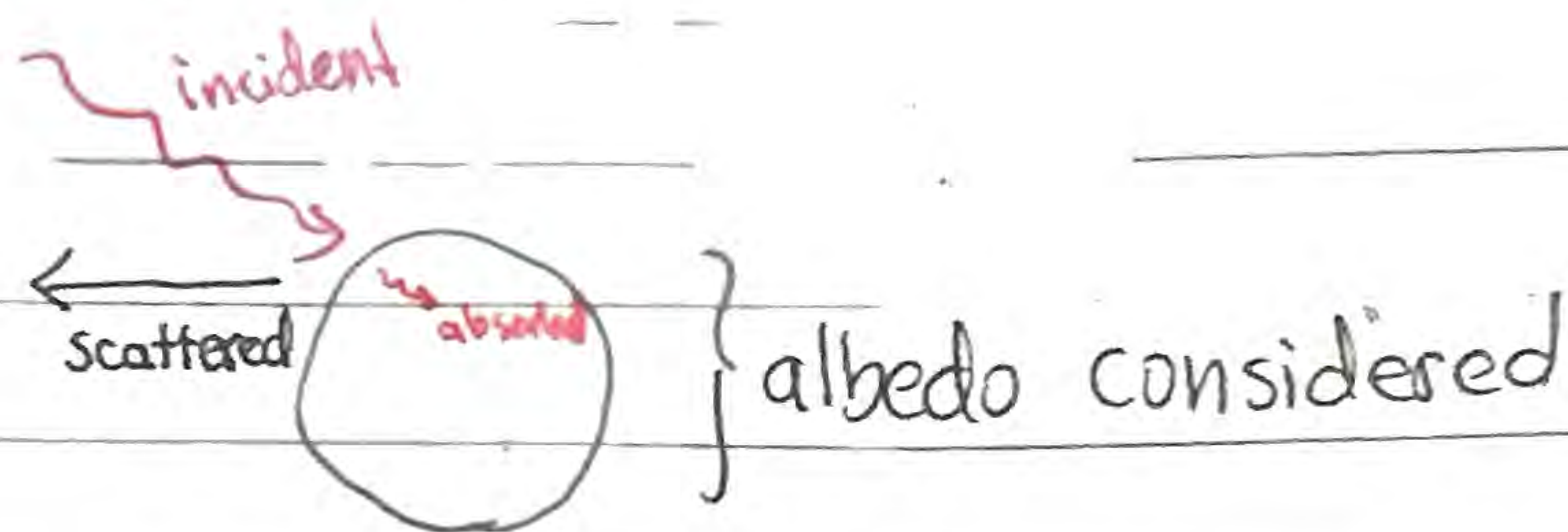
2. Emission of radiation from greenhouse gas molecules towards the planet surface



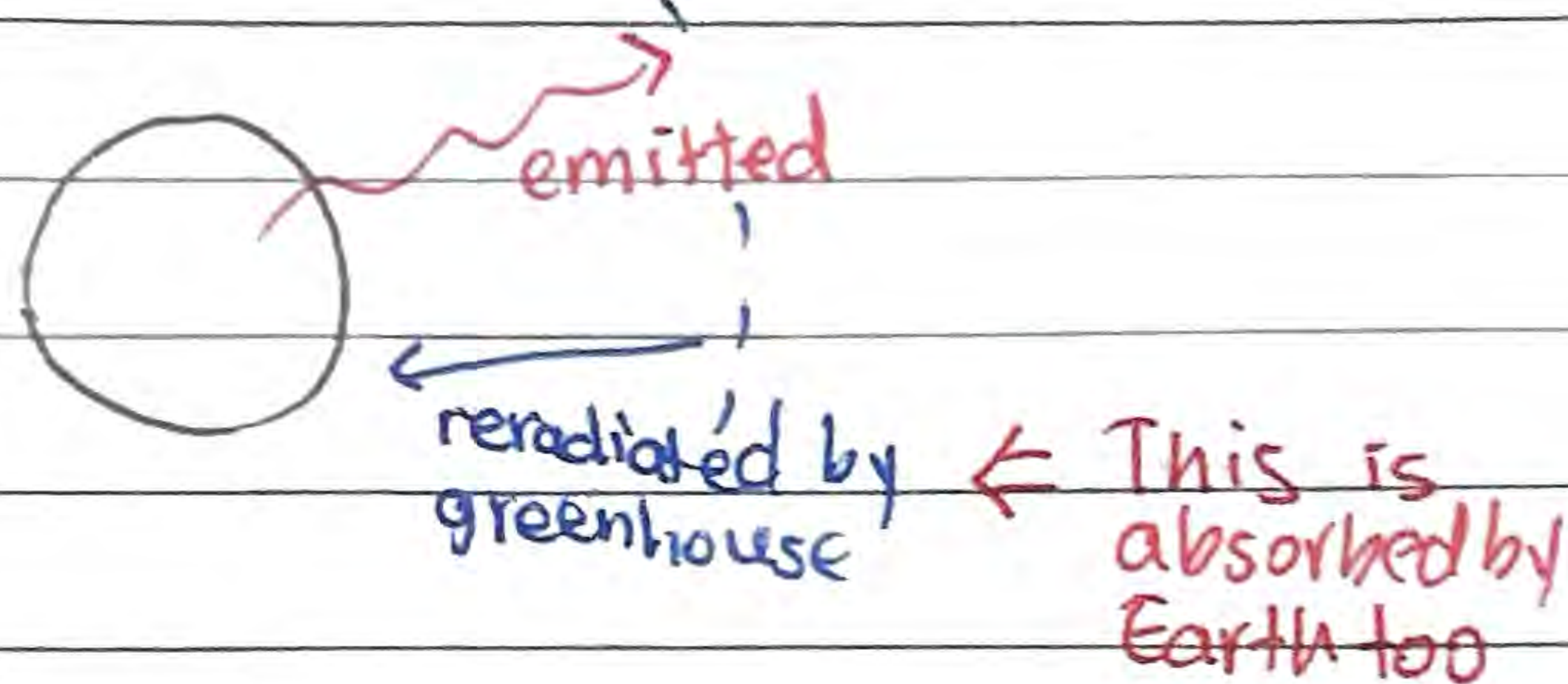
Greenhouse Effect

Conservation of Energy

All absorbed must be emitted



But then, the absorbed must be reradiated



Hence..

$$\text{Absorbed from incident} + \text{Reradiated by greenhouse} = \text{emitted by earth}$$